

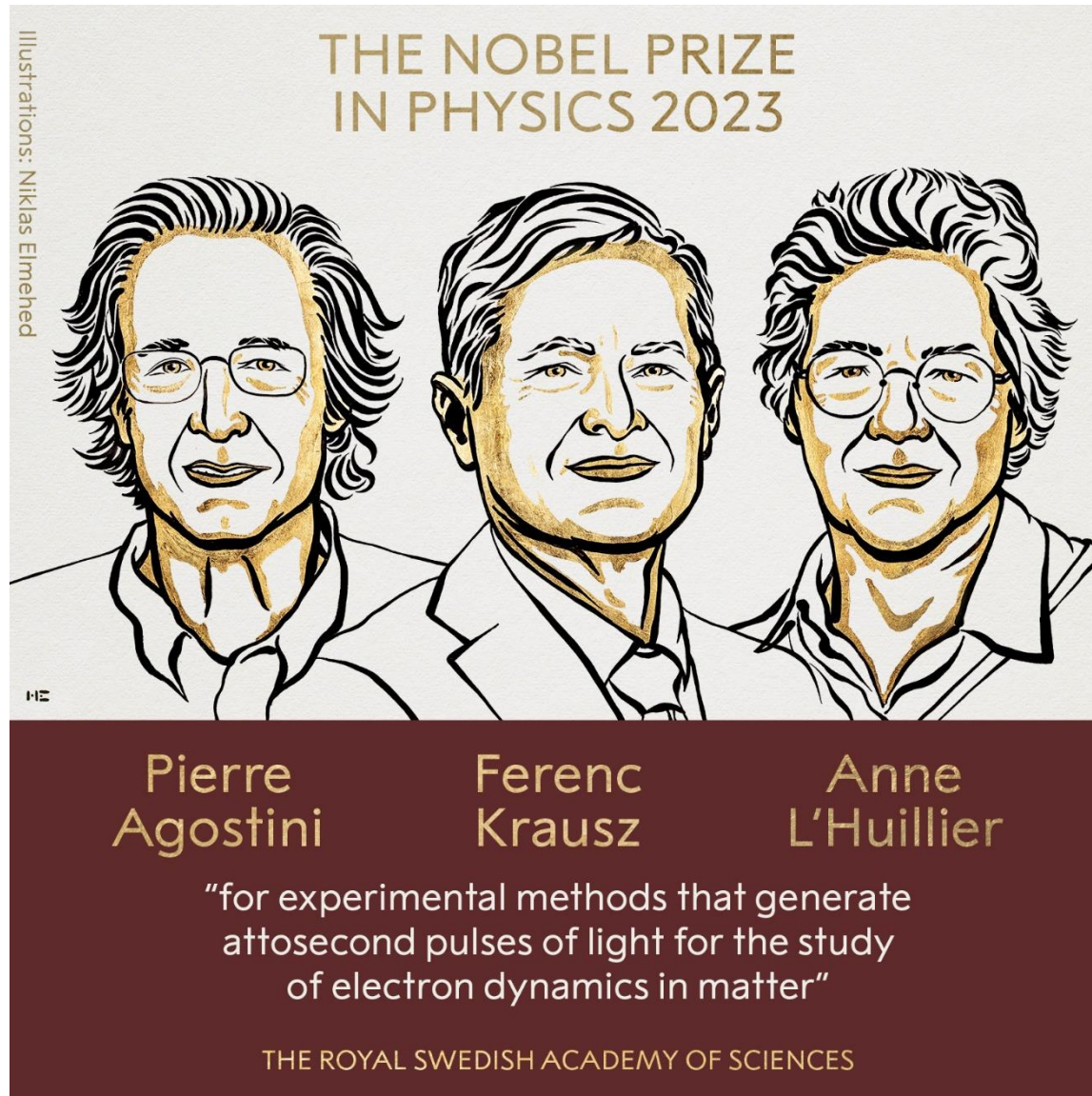


UE SPECTROSCOPIE AVANCÉE ET MATIÈRE DILUÉE

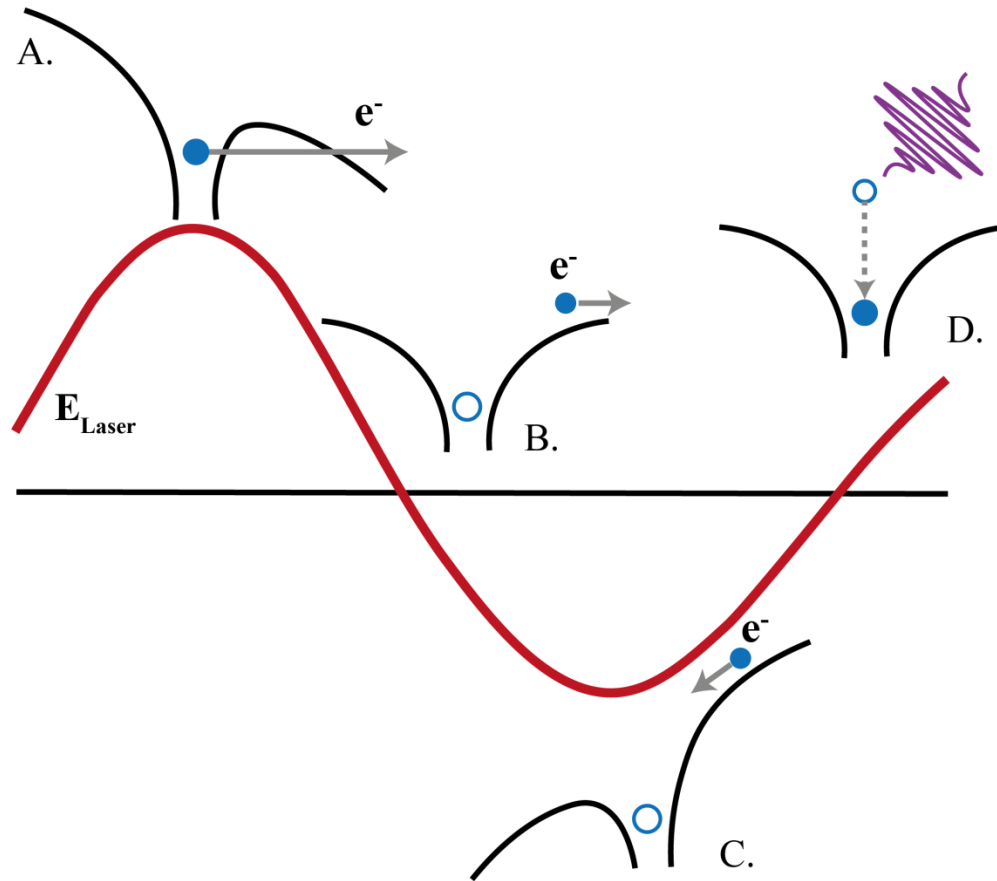
Photoelectron Spectra of noble gas

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High-order Harmonic Generation (HHG)

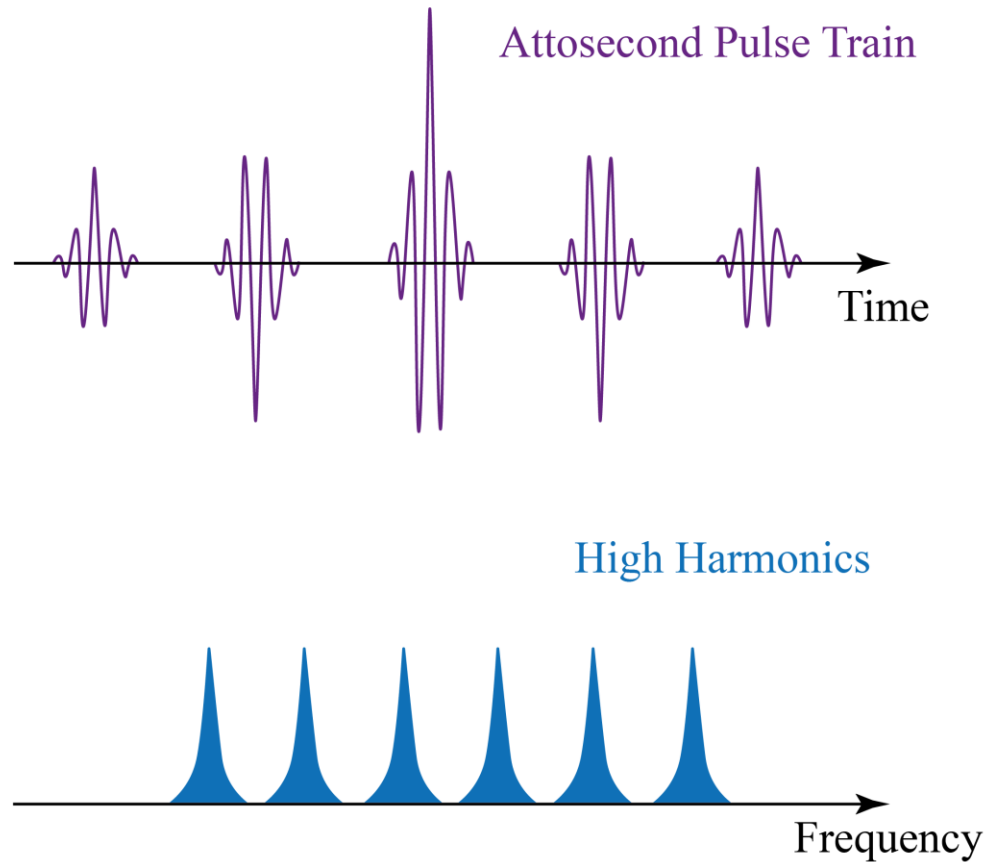


High-order Harmonic Generation (HHG)



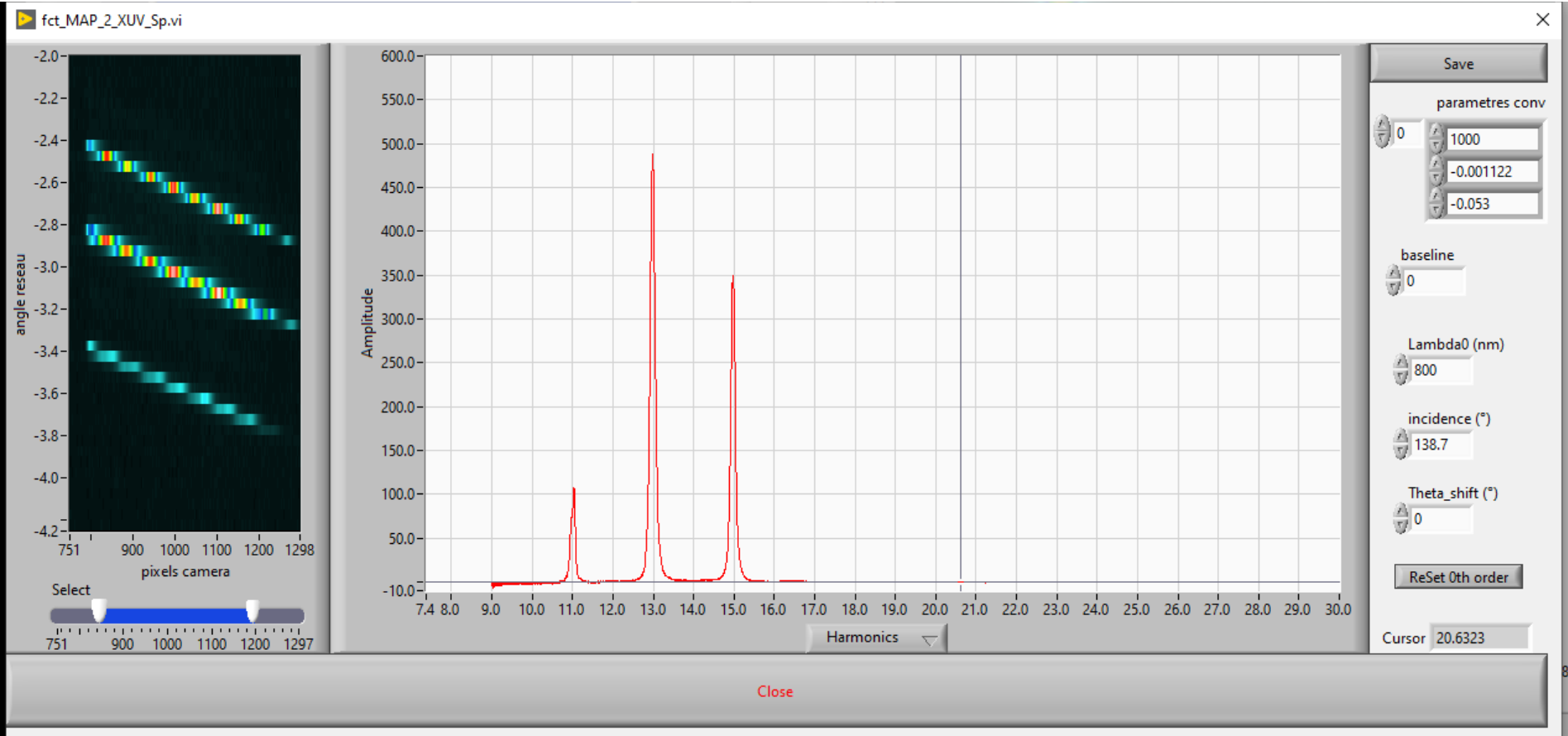
Three step model for HHG : Generation of attosecond pulses

High-order Harmonic Generation (HHG)



Three step model for HHG : Generation of attosecond pulses

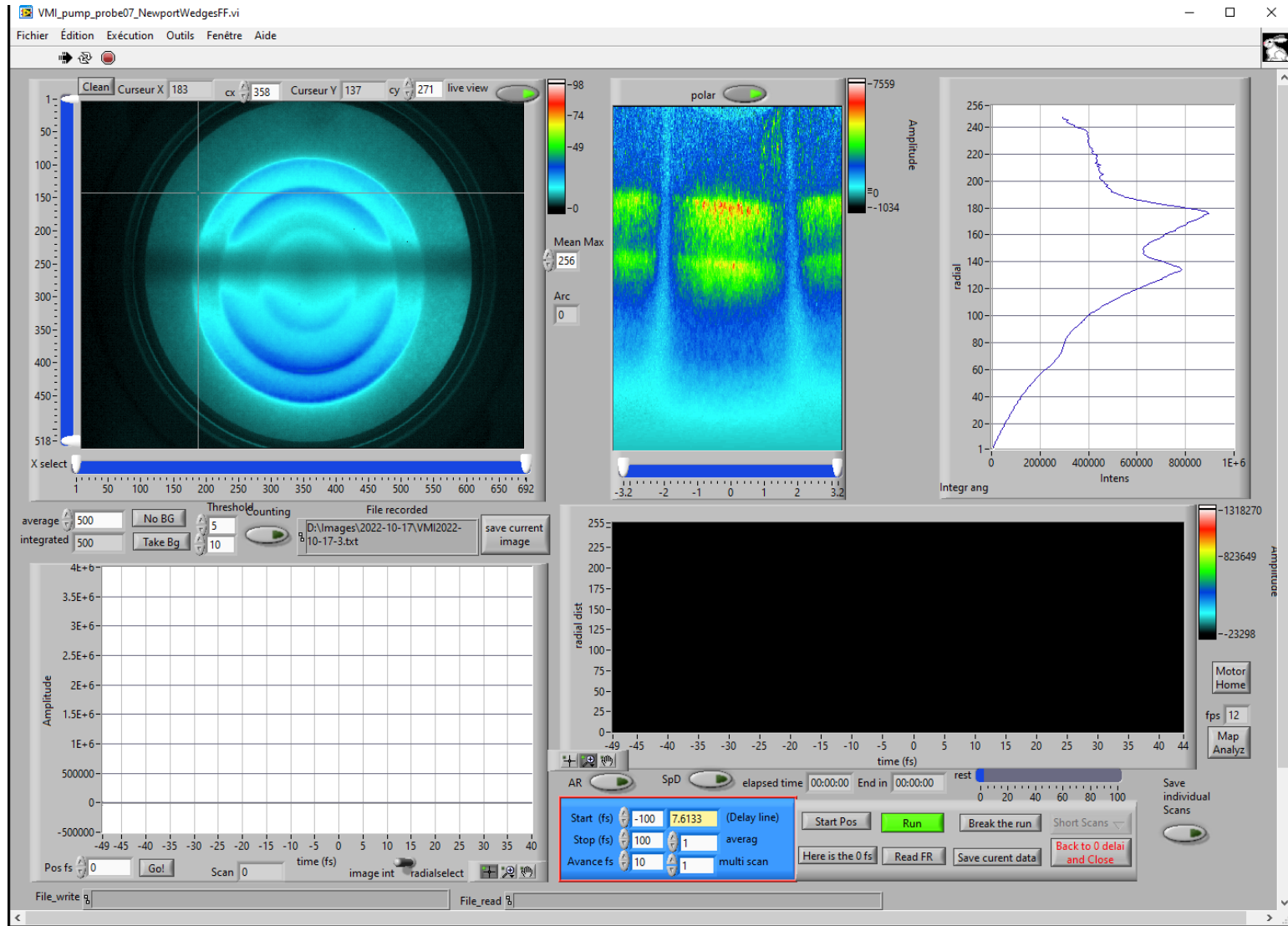
Photon Spectra from HHG



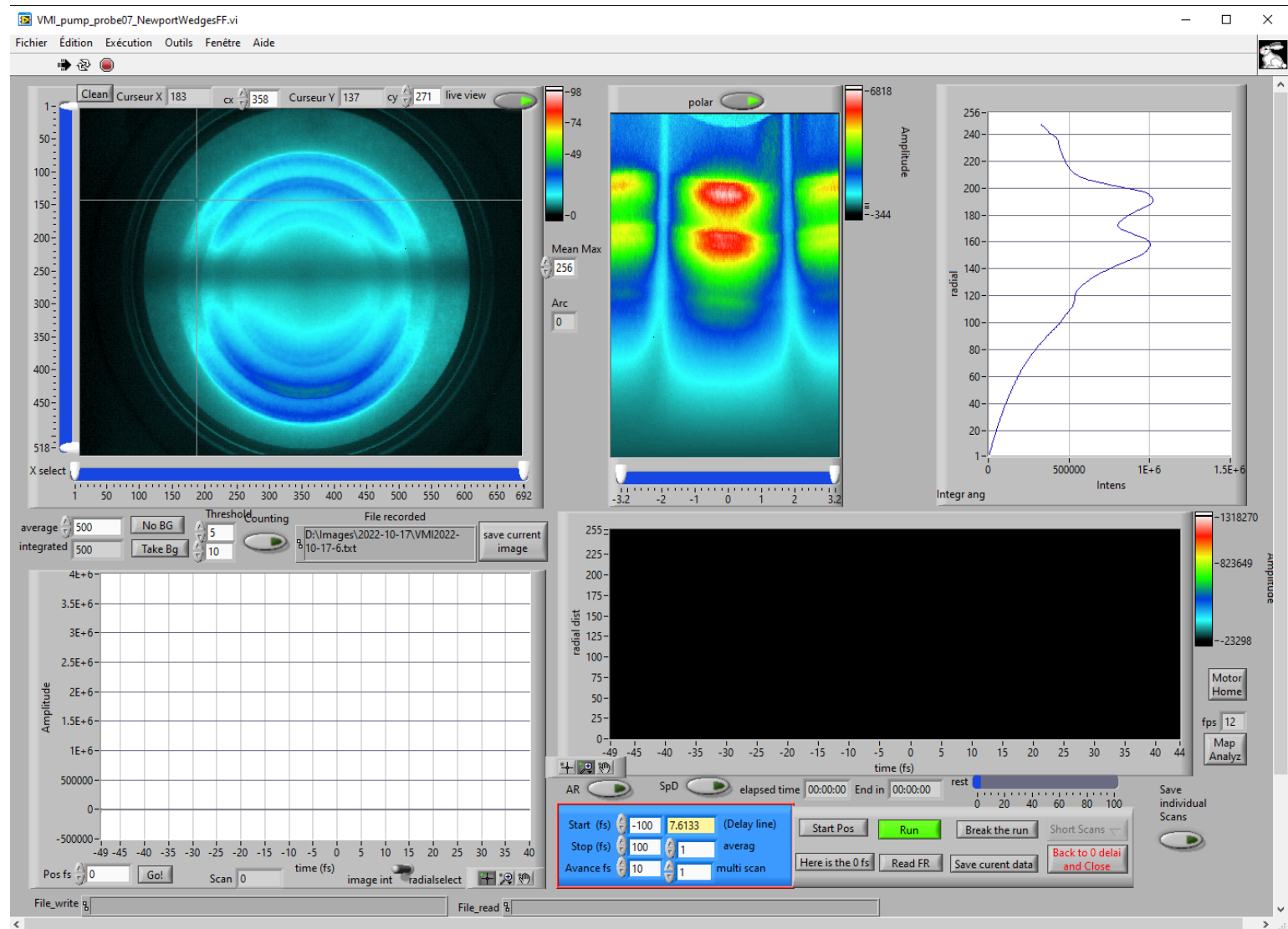
There are 3 high-order harmonics: **11, 13, 15**
What are the corresponding photon energies?

Use of HHG → Allows to vary photon energy, just like synchrotrons !

Photoelectron Spectra for Argon



Photoelectron Spectra for Krypton



How to extract information from photoelectron spectra?

Fit them with Gaussian function

$$g(x) = A \exp\left(-\frac{(x - b)^2}{2c^2}\right)$$

$$FWHM \approx 2.355 c$$

```
gnuplot> plot 'Krypton Photoelectron Spectra\Kr_PES_0.txt'
```

```
gnuplot> f(x)=g1(x)+g2(x)
```

```
gnuplot> g1(x)=p1*exp(-(x-m1)**2/(2*s**2))
```

```
gnuplot> g2(x)=p2*exp(-(x-m2)**2/(2*s**2))
```

```
gnuplot> m1=
```

```
gnuplot> m2=
```

```
.
```

```
.
```

```
.
```

```
gnuplot> fit [0:20] f(x) 'Krypton Photoelectron Spectra\Kr_PES_0.txt' via p1, m1, p2, m2, s
```

```
gnuplot> plot [14:20] 'Krypton Photoelectron Spectra\Kr_PES_0.txt' lc rgb 'blue', f(x) ls 1, g1(x) lc rgb 'black',  
g2(x) lc rgb 'green'
```